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PALAEOMAGNETISM OF THE MSISSI NORITE (MOROCCO) AND THE PALAEOZOIC RECONSTRUCTION OF GONDWANALAND

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Palaeomagnetic results are reported from eight sites in an Upper Devonian basic intrusion (the Msissi norite) in southeast Morocco. Specimens from one site are suspected of having been affected by lightning, but results from the other seven sites indicate the presence of a less-stable component, probably of viscous origin. The pole position corresponding to the stable component (0.5°S, 25°E, A_{95} = 16.5) is interposed between the Middle Cambrian/Ordovician pole and the Lower Carboniferous pole on the African polar wander curve When the southern continents are reassembled on the Smith/Hallam reconstruction of Gondwanaland the new Moroccan Devonian pole is in excellent agreement with the corresponding portion of the main Australian polar-wander curve. This places additional constraints on the possible date of fusion of the separate Southeast Australian plate with the rest of Gondwanaland, postulated recently on palaeomagnetic grounds by M.W. McElhinny and B.J.J. Embleton (1974). The combined African/Australian polar-wander curve is compared with the South American curve, and two possible interpretations of available data are discussed, one involving possible relative tectonic motion between South America and the rest of Gondwanaland during the Lower and Middle Palaeozoic, and the other, favoured here, requiring a reassessment of the ages of several South American pole determinations.

1. Introduction

Palaeozoic palaeomagnetic results are available from rocks of Cambrian to Ordovician, and Lower Carboniferous to Permian age from Africa (e.g. [1]), but hitherto no reliable study has been made on African rocks of Mid-Palaeozoic age. This paper describes the results of a recent study made on a basic plutonic intrusion in southeast Morocco, dated on the basis of stratigraphic evidence as Famenian (Upper Devonian) (Hollard, personal communication, 1969). The rock is a coarse-grained norite, consisting predominantly of calcic plagioclase and orthopyroxene, and outcropping over an area of some 5 km^2 in the northern Sahara, near the village of Msissi (location 30.93°N, 4.71°W). The adjacent and immediately overlying Devonian deposits appear to be horizontal and undisturbed, so that no tilt correction is considered necessary.

2. Sampling and measurement

Samples were collected by means of a portable electric coring drill driven from a mobile generator, and water for cooling and lubrication was supplied by an electric pump driven from the same generator. A total of eight sites were sampled at various parts of the intrusion, and between four and six separately orientated cores, 2.5 cm in diameter, and approximately 20 cm in length, were collected at each site. Orientation was performed by both sun-compass and magnetic compass, and the two values differed by less than two degrees at all sites except MB48, where they differed by up to seven degrees. A cylindrical sample, 2.5 cm in length, was subsequently cut from the lower part of each core in the laboratory. Measurements of the remanence of these samples were made on a PAR model SM2 magnetometer, and stability was tested by subjecting one pilot sample from each site to standard stepwise AF demagnetisation tests at 75-Oe intervals up to a maximum field of 900 Oe.

3. Results

With the exception of sites MB43 and MB48, the mean site NRM directions were distributed close to the present geomagnetic field direction at the collecting sites (Fig. 1a). Intensities of NRM ranged from 0.5 to 0.8 $\times$ 10^{-3} gauss at all sites except MB-48, at which the mean intensity was 10^{-1} gauss.

Pilot demagnetisation tests resulted in a movement of the remanence vectors from all sites except MB48 towards a steep northwest negative direction on treatment at 300–450 Oe, followed by random scattering of directions and erratic changes of intensity on further treatment up to 900 Oe (Fig. 2a). These results indicate the isolation of a stable component of remanence after treatment at 300–450 Oe, which at higher fields is either destroyed, or swamped by a spurious random moment introduced by the demagnetising apparatus. This was verified by subjecting all five samples from one site (MB49) to stepwise demagnetisation up to a maximum field of 600 Oe, and examining the dispersion of directions after each stage of treatment The results are shown in Fig 2b, in which each point on the stereogram represents the mean of the five individual sample directions after treatment at the value indicated, and the graph shows the circular standard deviation of these five directions plotted against the demagnetising field. The within-site dispersion of directions remains low (C.S.D. < 20°) on treatment up to 450 Oe, after which it increases rapidly, and the mean remanence vector after treatment at 300–450 Oe has a steep negative inclination and a northwesterly declination, in agreement with pilot samples from other sites. The remaining samples from the other sites were demagnetised at 300 Oe, and, excluding site MB48, this resulted in a reduction of between-site scatter from 28.5° to 11.5° (Fig. 1b and Table 1). In the case of site MB48 the direction of remanence of the pilot sample changed very little on treatment up to 750 Oe, and the mean site direction after treatment at 300 Oe remained remote from the other seven sites.

4. Discussion

The close proximity of six of the eight mean site NRM directions to the present geomagnetic field direction at Msissi suggests the presence of a secondary component of remanence of recent origin in the NRM of these samples. The good grouping of seven of the mean site vectors around a steep negative northwest direction after AF treatment indicates the successful removal of this secondary component, and the isolation of a stable underlying component, probably of primary origin. In the case of site MB48 the initial high intensity of the magnetisation (two orders of magnitude higher than the other sites), and the large

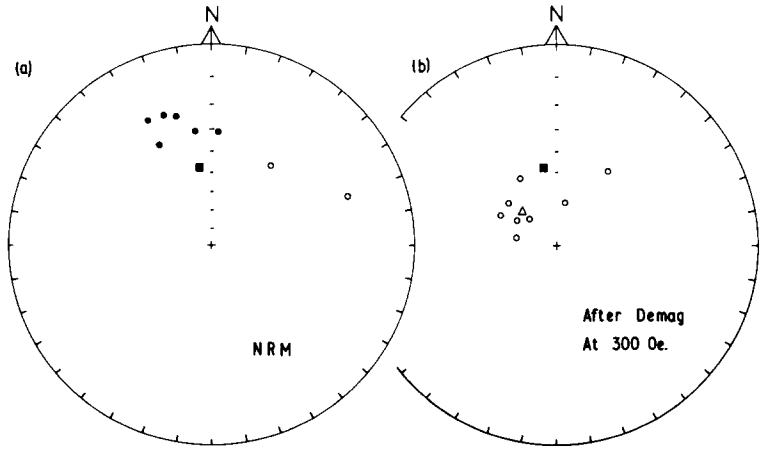


Fig. 1. Mean site directions of magnetisation plotted on stereographic projections. Solid symbols indicate lower hemisphere, open symbols indicate upper hemisphere. a. NRM directions. b. After AF demagnetisation at 300 Oe. Black square denotes present geomagnetic field direction at sampling locality.

TABLE 1

Mean site directions of magnetisation, palaeomagnetic pole positions and statistical parameters

Site No.	NRM						After treatment at 300 Oe					Palaeomagnetic south pole		
	D	I	N	R	K	α_{95}	D	I	R	K	α_{95}	lat.	long.	A_{95}
MB42	3.5	+31.5	4	3.745	12	28	302.5	−64.0	3.984	193	6.5	3.0S	211.5E	10
MB43	70.5	−19.0	6	5.995	1044	2	11.5	−65.5	5.905	52	9.5	10.0N	168.5E	13
MB44	345.0	+23.0	5	4.966	117	7	312.0	−55.0	4.927	55	10.5	9.5S	213.0E	14
MB45	332.5	+31.5	6	4.907	5	35	281.0	−67.5	5.603	13	19.5	16.5S	214.5E	31
MB46	340.0	+21.0	6	5.995	1025	2	298.5	−55.0	5.978	230	4.5	2.0N	221.0E	6
MB47	352.0	+30.5	6	5.982	276	4	331.5	−48.5	5.966	148	5.5	23.5N	202.0E	7
MB48	37.0	−38.0	4	3.985	200	6.5	35.0	−41.5	3.995	596	4	25.5N	139.5E	4
MB49	333.5	+20.5	5	4.749	17	19.5	314.5	−69.0	4.935	61	10	3.5S	200.5E	16
Overall mean	359.5	+16.0	8	6.249	4	31.5	328.5	−64.0	7.488	14	15.5	4.0S	18.0E	22
Overall mean excl. MB48	354.0	+23.5	7	5.887	5	28.5	315.5	−63.0	6.797	29	11.5	0.5S	25.0E	16.5

D = declination (E of N) in degrees, I = inclination (positive downwards) in degrees, N = number of samples averaged, R = length of vector resultant of N unit remanence vectors, K = precision parameter, α_{95} = radius of 95% confidence circle about mean direction, A_{95} = radius of 95% confidence circle about mean pole.

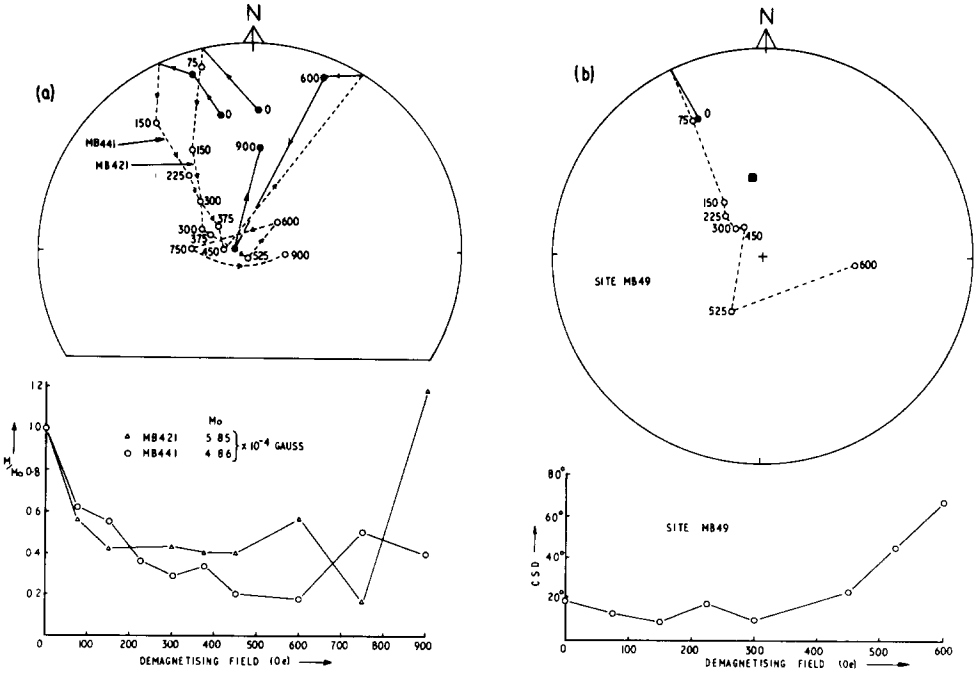


Fig. 2. a. Examples of typical pilot-sample behaviour during stepwise demagnetisation. Notation as in Fig. 1 b. Results of subjecting all five samples from site MB49 to stepwise demagnetisation Each pole represents the mean of the five separate directions after each stage of treatment. Numbers indicate peak field in oersteds. The graph shows the standard deviation of these five directions plotted against the demagnetising field.

difference between the magnetic and sun-compass orientation values, indicates the presence of a strong local magnetic field, and it is possible that rocks at this site may have been affected a lightning strike

Graham [2] has observed that anhysteretic magnetisations produced by lightning strikes are generally magnetically "soft", and can often be effectively removed by AF demagnetisation. However in specimens in which the anhysteretic magnetisation is more than about 10 times stronger than the primary thermal remanence, AF demagnetisation does not effectively restore the original direction of magnetisation. This appears to be the case for site MB48, in which the mean direction of remanence remains removed from that of the other sites, even after AF treatment to 750 Oe. Consequently results from this site have not been included in the final statistical analysis.

The mean direction of stable remanence obtained from the other seven sites after AF cleaning is thought to represent a good approximation to the direction of the geomagnetic field with respect to Northwest Africa during the Upper Devonian. The sampling sites were evenly distributed over the total area of exposure of this formation, and since the coarse grain size indicates a slow rate of cooling, it is probable that these results at least partially average out the effects of secular variation of the geomagnetic field.

5. The African Palaeozoic polar-wander curve

The mean pole position derived from the Msissi norite after AF cleaning (pole 10) is plotted, together with other African Palaeozoic pole positions in Fig. 3, against a background of the Smith-Hallam reconstruc-

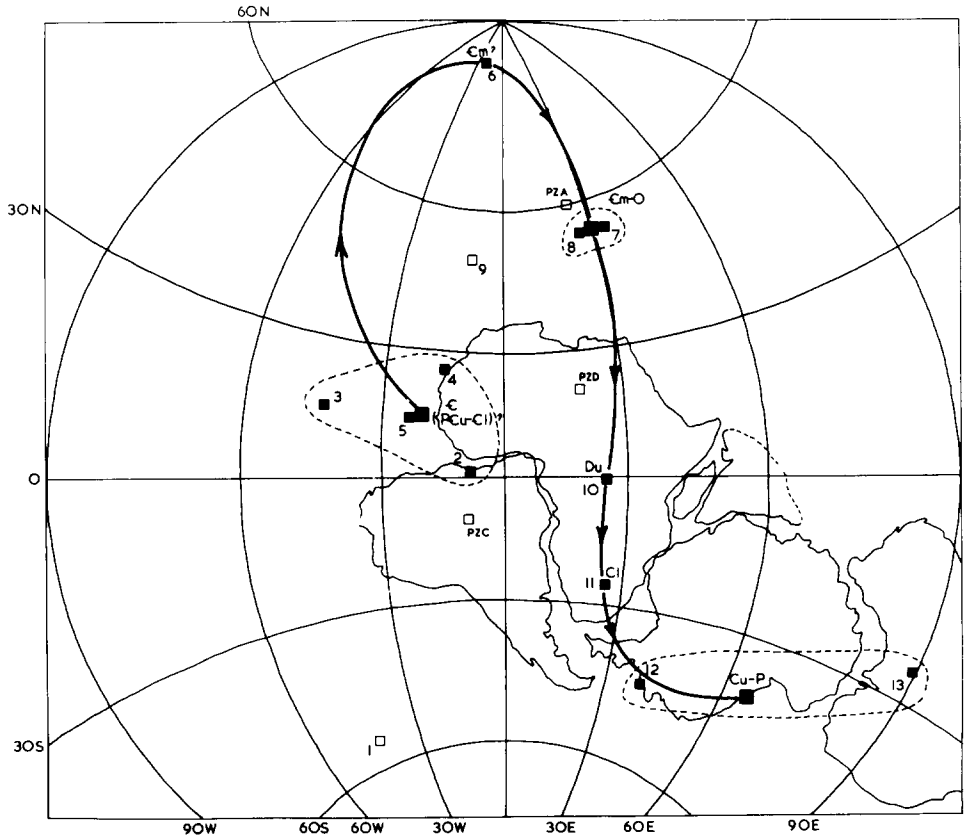


Fig. 3. African Palaeozoic polar-wander curve. Small solid symbols denote poles based on AF or thermal stability tests and cleaning procedures, small open symbols denote poles for which insufficient data has yet been published to allow reliable assessment of validity, and large solid symbols denote mean poles, calculated for successive age groups. Numbers and symbols refer to Table 2.

TABLE 2

Palaeozoic pole positions

Pole No.	Age	Formation	Ref.	Formation pole		Mean pole	
				lat.	long.	lat.	long.
(A) *Africa*							
1	Є1 (?)	Fish River Series	[11]	55N	137E		
2	PЄ–Є	Sijarira Group	[12]	2N	8W		
3	PЄ–Є1	Klipheuvel Formation	[13]	16N	44W	15N	23W
4	PЄ(630 ± 24)	Ntonya Ring Structure	[15]	28N	15W		
5	01(500 ± 17)	Hook intrusives	[6]	14N	24W		
6	(540 ± 25)?	Sabaloka Ring Structure	[9]	83N	21W		
7	Єm	Moroccan lavas	[16]	53N	34E	53N	30E
8	Є–O	Tassili sediments	[17]	53N	26E		
9	O	Table Mt. Series	[18]	50N	11W		
10	Du	Msissi norite	This paper	1S	25E		
11	Cl	Dwyka varves	[19]	26S	26E		
12	Cu (?)	Galula redbeds	[20]	45S	40E	40S	64E
13	Pl (?)	Ecca redbeds	[20]	27S	89E		
PZA	< 1000	Plateau Series Zambia (1)	[21]	71S	7W		
PZB	< 1000	Plateau Series Zambia (2)	[21]	60N	25E		
PZC	L. Pal	Plateau Series Zambia (3)	[21]	10S	8W		
PZD	L. Pal	Plateau Series Zambia (4)	[21]	22N	19E		
(B) *Australia*							
1	Є1	Antrim Plateau volcanics	[22]	9N	160E	0N	28W
2	Є1	Arumbera s.st.	[23]	8N	35W		
3	Єm	Hudson formation	[24]	18N	19E	15N	28E
4	Єm	Hugh River shale	[23]	11N	37E		
5	Om	Stairway s.st.	[25]	2S	50E	8S	38E
6	Ol	Jinduckin fm.	[24]	13S	25E		
7	S?-D	Mereenie s.st.	[25]	41S	40E		
8	Cl	Gilmore and Isismurra volcanics	[26]	73S	146W	76S	180E
9	Cu(298)	Paterson toscanite	[27]	73S	147E		
10	Cu	Main glacial stage	[27]	53S	149E		
11	Cu	Rocky Creek conglom	[27]	51S	138E		
12	Cu	Currabubella fm.	[27]	42S	135E	46S	135E
13	P(l)	L. marine basalt/Moonbi lamprophyre	[27]	40S	125E		
14	Pm–u(248)	U. marine latites	[27]	44S	132E		
F.R.S.	Єm	Lower Lake Frome Group (Flinders Ranges)	[5]	8S	25E		
P.Q.	PЄ(latest)	Pound quartzite	[5]	60S	6E		
B.C.	Cu	Upper Lake Frome Group (Balcoracana Creek)	[5]	38S	26E		
A.D.	> Є1	Aroona Dam sediments	[5]	36S	33E		
Southeast Australia (Pre-Carboniferous)							
15	Sm–u	Duaro and Laidlaw volcanics	[26]	54N	91E		
16	Su(400)	Mugga Mugga porphyry	[28]	80S	20W		
17	Dl	Bowning volcanics	[26]	64S	45E		
18	Dl	Ainslie volcanics	[26]	71S	7W		
19	Du	Lochiel formation	[26]	58S	40W		

TABLE 2 (continued)

Pole No.	Age	Formation	Ref.	Formation pole		Mean pole	
				lat.	long.	lat.	long.
(c) *South America*							
1	Є	Purmamarca Village	[29]	61N	67W		
2	Є	South Tilcara	[29]	52N	27E	50N	25E
3	Є	North Tilcara	[29]	49N	24E		
4	Є	Abra de Cajas	[29]	2N	28E		
5	Є	Purmamarca	[29]	5N	39E	13N	27E
6	O	Salta sediments	[29]	31N	13E		
7	Є–O	Red beds, Salta and Jujuy	[30]	15N	22W		
8	Є–O	Red beds, Salta and Jujuy	[30]	8N	40W	13N	25W
9	O–S(?)	Urucum fm., Brazil	[30]	17N	13W		
10	O	Bolivian sediments	[30]	4N	58W	5N	56W
11	Dl (O?)	Bolivian sediments	[31]	7N	53W		
12	Dl	Picos and Passagem series, Brazil	[31]	30S	47W		
13	Cu	Piaui fm., Brazil	[31]	50S	15W		
14	P	Red beds, Argentina	[32]	65S	13W		
15	C	Bum Bum fm., Argentina	[33]	85S	25W		
16	Pl	La Colina basalt	[33]	66S	12W	70S	19W
17	P	Los Colorados (lowest beds)	[33]	69S	25W		
18	P–C	Huaco gorge	[34]	68S	6W		
19	P	Los Colorados (middle beds)	[33]	74S	47W		
20	P	La Colina sediments	[33]	81S	33W		
(D) *India*							
1	PЄ–Є	Bhander s.st.	[35]	49S	33E		
2	PЄ–Є	Upper Rewa s.st.	[35]	35S	42E		
3	PЄ–Є	Uper Bhander s.st.	[35]	32S	19E	34N	32E
4	Єl	Salt Range s.st.	[36]	28S	32E		
5	Єm	Salt pseudomorphs beds	[37]	27S	33E		
6	Cu	Talchir series	[38]	32S	46W	26S	48W
7	Pu	Kamthi beds	[39]	21S	50W		
(E) *Antarctica*							
1	Cu(502)	Mirnyy charnockites	[40]	2S	28E	15S	19E
2	Ol(480)	Sør Rondane intrusives	[41]	28S	10E		

tion of Gondwanaland [3]. The shape of the Lower Palaeozoic part of the polar-wander curve is currently under dispute [4,5]. The position of the pole off the coast of Northwest Africa during Late Precambrian to Lower Cambrian times is now fairly well established, on the basis of the group of four palaeomagnetic poles in this region (poles 2–5, Fig.3 and Table 2A). The radiometric age determination on the Ntonya Ring Structure (pole 4) of 630 ± 24 m.y. is perhaps representative of this group, and although the gabbros and diorites of the Hook intrusives (pole 5) are often regarded as Ordovician, this age is based on a radiometric determination on an associated granite, which is not necessarily applicable to the basic bodies as well [6]. Pole 9, derived from the (Ordovician?) Lower Table Mountain Series shales of Cape Town is of dubious reliability , since it is based on only eight cores from a 2 ft thick red siltstone band, and no details of demagnetisation stability tests are available [7].

A second group of poles off the north coast of Africa (poles 7 and 8) is of Middle Cambrian to Ordovician age, and thus younger than the first group, whereas the Sabaloka Ring complex (pole 6) has yielded a radiometric age of 540 ± 25 m.y. [8] which places it intermediate in age between these two groups of poles. In view of these age relationships, it is proposed that the Palaeozoic polar-wander path for Africa follows a clockwise loop, as indicated in Fig. 3. This interpretation differs somewhat from that of Piper et al. [4], and Briden [9], but is similar to that of McElhinny et al. [1].

The Devonian pole position from the Msissi norite described in this paper then lies on an almost straight meridional path between the mean Middle Cambrian/Ordovician pole, and the Lower Carboniferous pole from the Dwyka varves (Fig. 3). The relative ages of the Galula and Ecca redbeds (poles 12 and 13) have been discussed elsewhere (e.g. [1]), and in the absence of confirmatory data, the polar-wander path is drawn through the mean of these two poles, but passing near the Galula pole (Upper Carboniferous?).

6. The Australian Palaeozoic polar-wander curve.

Australian palaeomagnetic data has recently been reviewed by McElhinny and Embleton[10], who suggest that Southeast Australia formed part of a separate plate from the rest of Australia during the Lower Palaeozoic, and that the two regions became welded together in Devonian times. Reliable Australian Palaeozoic pole positions are listed in Table 2B, and groups of poles of successive ages have been used to calculate a series of mean poles, plotted as large solid circles on a Smith-Hallam computer reconstruction in Fig. 4. Poles 15–19 were derived from Southeast Australia, and appear to lie on a separate and

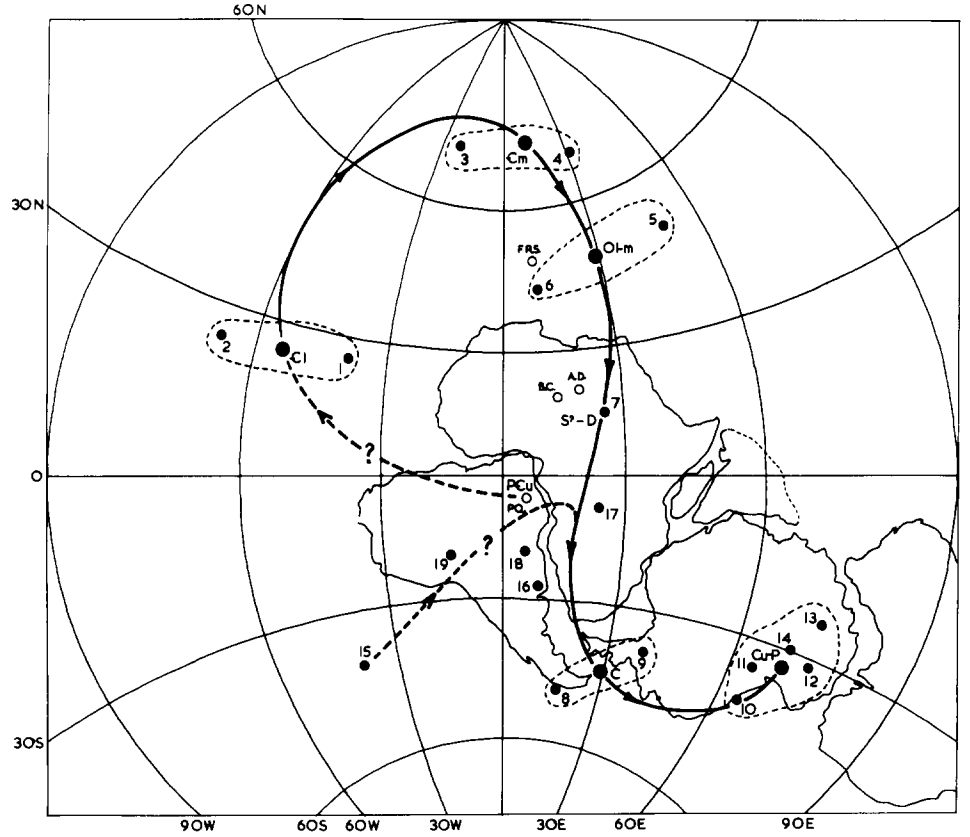


Fig. 4. Australian Palaeozoic polar-wander curve plotted on the computer reconstruction of Gondwanaland of Smith and Hallam [3]. Symbols as in Fig. 3. Note the separate branch for Southeast Australian poles, uniting with the main curve in Mid-Palaeozoic times.

distinct polar-wander curve, that unites with the main Australian curve in the vicinity of the Devonian pole positions. Poles from the Gilmore and Isismurra volcanics, and the Paterson toscanite, of Carboniferous age (poles 8 and 9, respectively) appear to be distinct from the main group of Upper Carboniferous to Permian poles (10–14), and accordingly two separate mean poles have been calculated, and used to define the Upper Palaeozoic part of the Australian polar-wander curve. McElhinny and Embleton [10] note that difference between these two groups of poles may reflect local tectonic disturbance, but the good agreement with the African polar-wander curve (see next section) suggests that it may indeed represent a true apparent polarshift.

7. Superimposition of the African and Australian polar-wander curves

The African and Australian polar-wander curves, based on the successive groups of poles discussed above, are shown superimposed on a Smith-Hallam computer reconstruction of the pre-drift positions of the southern continents in Fig. 5. The agreement between these two curves is remarkably good, and lends strong support to the validity of this reconstruction of the relative positions of Africa and Australia during Palaeozoic times. The new Moroccan Upper Devonian pole position is situated on the combined African/main Australian curve *before* the postulated junction with the Southeast Australian curve (Figs. 4

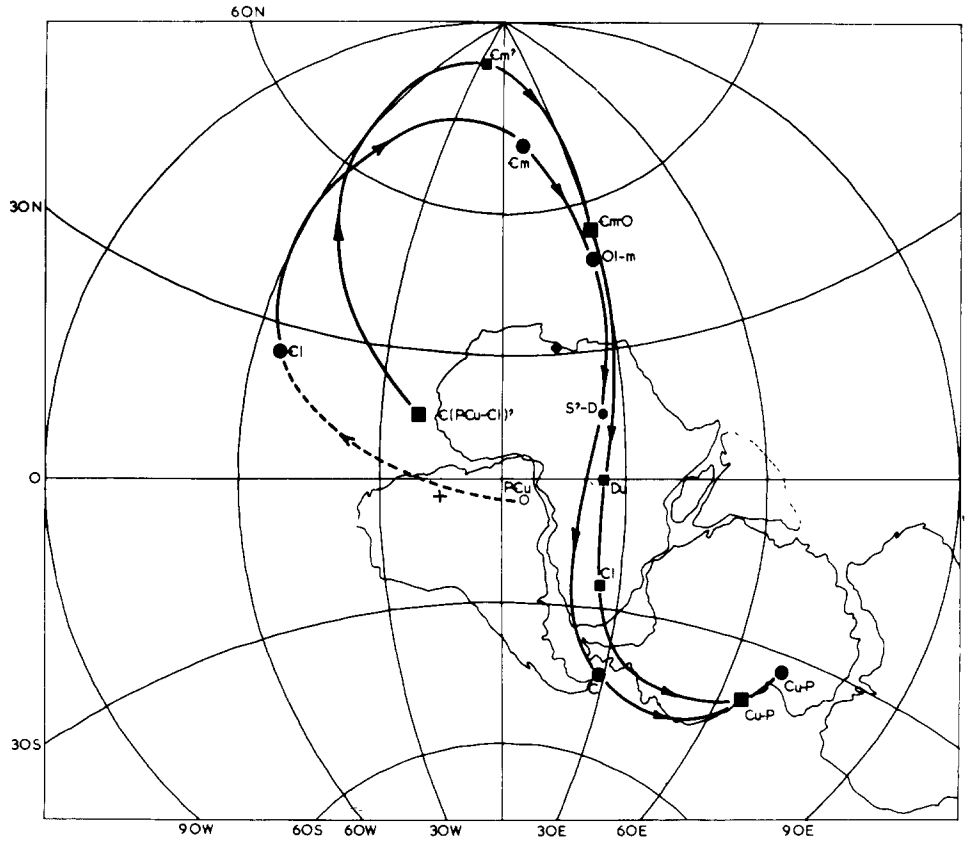


Fig. 5. Superimposition of African and Australian polar-wander curves on the computer reconstruction of Smith and Hallam [3] Also shown are the mean Late Precambrian to Mid Cambrian pole for India (cross) and the mean Upper Cambrian/Lower Ordovician pole for Antarctica (diamond) – see Table 2

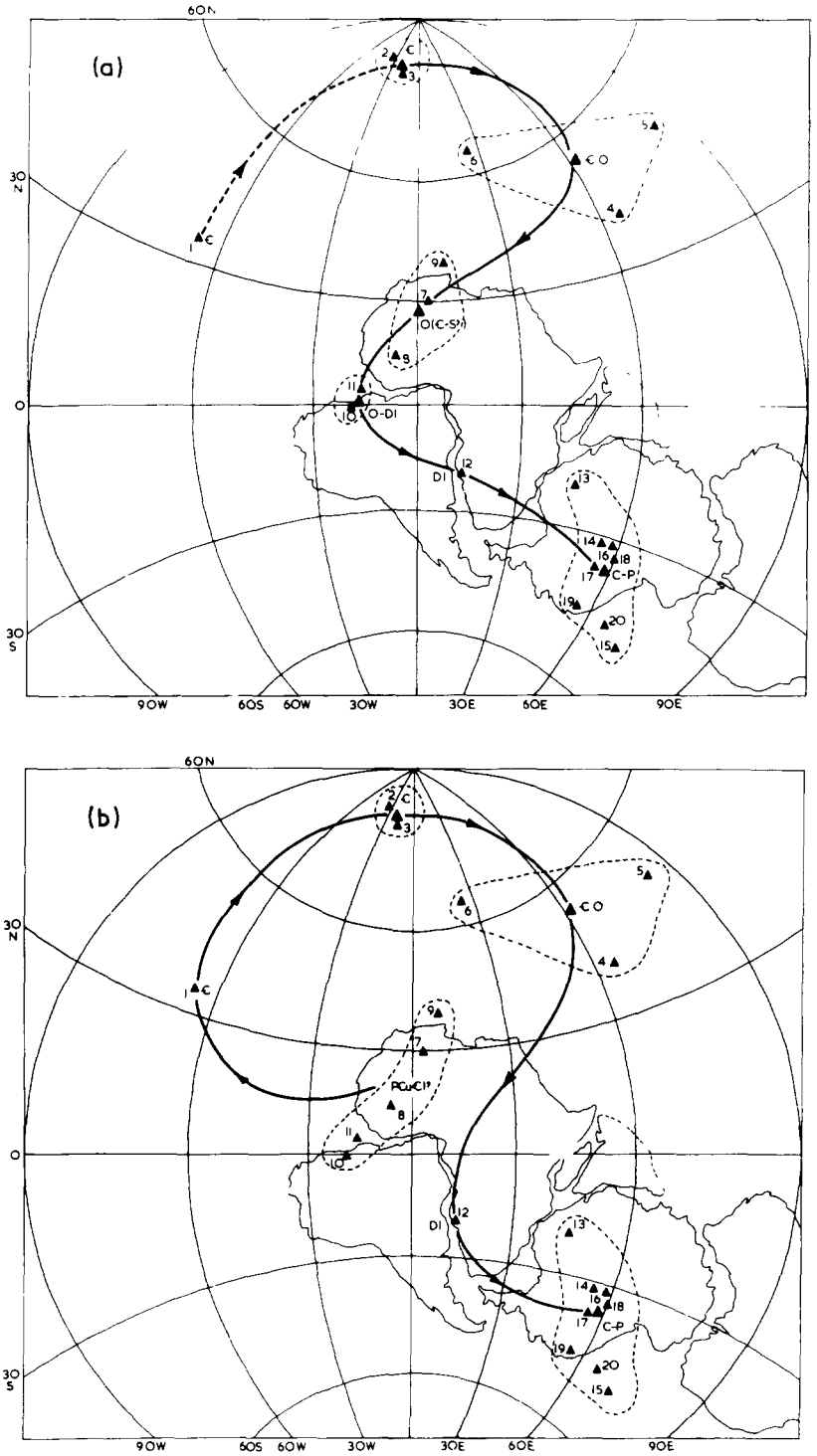


Fig. 6. Two possible interpretations of South American palaeomagnetic data in terms of polar-wander curves. In (a), the smoothest possible curve has been drawn through the successive mean poles, based on the currently accepted ages of the relevant formations In (b) it is assumed that the group of poles in the vicinity of Northwest Africa are all of Late Precambrian to Lower Cambrian age, so that the shape of the polar-wander curve becomes similar to those of Africa and Australia (Fig. 5).

and 5). This places an additional constraint on the possible time at which the two parts of Australia became united, and indicates that this event probably occurred in *post-Famenian* times

8. The South American polar-wander curve

All available reliable Palaeozoic pole positions for South America, derived from studies involving AF or thermal cleaning procedures, are summarised in Table 2C, together with mean pole positions calculated from successive groups of poles There appear to be two possible interpretations of this data in terms of polar-wander curves for South America. In Fig. 6a the smoothest possible curve has been drawn through the successive mean poles, based on the currently accepted ages of the relevant formations. This curve differs from the combined African/Australian curve in two important respects. Firstly there appears to be no evidence for a Late Precambrian/Lower Cambrian pole position off the coast of Northwest Africa; and secondly the South American curve displays a pronounced swing to the west, across Northwest Africa and Brazil during Mid-Palaeozoic times, which is not apparent on either the Australian or the African curves. Taken at its face value, this suggests the possible occurrence of significant relative tectonic motion between South America and the rest of Gondwanaland in Palaeozoic times, for which as yet no geological evidence has been put forward.

An alternative possible interpretation of South American Lower and Middle Palaeozoic poles, recently proposed by McElhinny et al. [5], is illustrated in Fig. 6b. In this case it is assumed that the group of South American poles in the vicinity of Northwest Africa (poles 7–11) are all of Late Precambrian to Lower Cambrian age, so that a nearly-closed clockwise loop is present in the Lower Palaeozoic part of the curve, as in the African and Australian curves. This interpretation is in much better agreement with the new African Devonian pole described in this paper, and the polar-wander curves from the three continents can then be superimposed over their entire lengths, within the error limits of the respective palaeomagnetic poles, when plotted on the Smith-Hallam pre-drift reconstruction. Thus it no longer becomes necessary to postulate significant Palaeozoic relative motion between South America and the rest of Gondwanaland. Available Lower Palaeozoic Indian and Antarctic poles also lie close to this common polar-wander curve (Fig. 5), providing further evidence for the unity of Gondwanaland throughout this time. Clearly a reassessment of the ages of the five South American sedimentary formations concerned, or else further studies on well-dated Late Precambrian to Lower Cambrian South American rocks is necessary before a final resolution of this problem will be possible.

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